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RESEARCH and PROFESSIONAL EXPERIENCE

University of Illinois Chicago (2022 – Present)

- **Warren S. McCulloch Professor**, *Department of Computer Science* (with courtesy appointment in Electrical and Computer Engineering) (2024 – Present). Tenured faculty member engaged in research and instruction and a member of the Honors College. Research spans scientific visualization, large-scale data analytics, and the design of research infrastructure for scientific discovery. Ongoing efforts address the analysis of data produced by supercomputers, large-scale scientific instruments, and sensor networks, with the goal of integrating edge and high-performance computing across the computing continuum.
 - Professor (2022 – Present)
 - Visiting Professor (2022)
- **Director**, *Electronic Visualization Laboratory* (2024 - Present), Leads an interdisciplinary laboratory that performs research in the areas of high-performance visualization, virtual reality, and networked collaboration. EVL is home to pioneering technologies such as CAVE, SAGE, and CAVE2. Facilitates global collaborations and advances in visual data science, pushing the boundaries of big data challenges and high-resolution visualization.
- **Director**, *Software Technologies Research Center (Softech)* (2024 – Present), Oversees the research center responsible for licensing and commercialization of groundbreaking technologies developed by the Electronic Visualization Laboratory.
- **Co-Chair**, *CS + Design Program* (2024 – Present), Oversees the computer science component of the *Computer Science + Design* program. This undergraduate program enables students to acquire computational skills while applying them creatively to design projects. Collaborates with design faculty, coordinates interdisciplinary coursework, and supports students in developing innovative solutions that blend technology and design.

Argonne National Laboratory (1992 – Present)

- **Argonne Distinguished Fellow** (2025 – Present), Recognized for exceptional leadership and research in high-performance computing, large-scale data analytics, and advanced visualization.
- **Founding Co-Director**, *George Crabtree Institute for Discovery* (2024 – Present), a joint institute between the University of Illinois Chicago and Argonne National Laboratory. In collaboration with the University of Illinois Chicago co-director, responsible for the overall direction of the institute, including setting its strategic focus and developing joint research efforts between the two institutions.
- **Division Director**, *Argonne Leadership Computing Facility* (2010 – Present), Responsible for strategic direction of the facility, which includes planning and executing facility upgrades (in collaboration with the project director) and integrated data strategies, risk assessment and management, operations reporting and reviews, staff promotion and retention, and safety. The ALCF is home to some of the

world's fastest supercomputers: *Polaris*, an HPE Apollo 6500 Gen10+ supercomputer, and *Aurora*, an Intel CPU/GPU supercomputer currently ranked as one of the fastest AI (HPL MxP) systems in the world that supports scientific and engineering open science research.

- **Deputy Associate Laboratory Director**, *Computing, Environment and Life Sciences Directorate* (2006 – Present), Provides strategic planning and management of the Laboratory Directorate comprising Biosciences, Computational Science, Data Science and Learning, Environmental Sciences, Mathematics and Computer Science divisions, and the Argonne Leadership Computing Facility. Executes a research program that couples computing-related activities with various science domains whose future depends on progress in computing.
- **Research Scientist** (1992 – Present), Active research activities in high-performance computing, scientific visualization of large and complex datasets, information visualization related to high-performance computing, virtual reality for science, and the integration of edge services.
 - Senior Scientist - RD6 (2012 – Present)
 - Scientist - RD5 (2004 – 2012)
 - Scientist - RD4 (2001 – 2004)
 - Scientist - RD3 (2000 – 2001)
 - Assistant Scientist - RD2 (1996 – 2000)
 - Scientist Associate - RD1 (1992 – 1996)

University of Chicago (1999 – Present)

- **Senior Scientist**, *The University of Chicago Consortium for Advanced Science and Engineering* (2018 – Present), Member of an organization focused on facilitating research collaborations between faculty at UChicago and scientists and scholars at other institutions, including Argonne, Fermilab, and Marine Biological Laboratory.
- **Senior Fellow**, *Computation Institute, a joint institute of the University of Chicago and Argonne National Laboratory*, Member of a multi-disciplinary research center that worked to advance complex, system-level scientific problems using a wide range of computational approaches and computing platforms.
 - Senior Fellow (2004 – 2018)
 - Fellow (1999 – 2004)

Northern Illinois University (2012 – 2023)

- **Presidential Research, Scholarship and Artistry Professor**, *Department of Computer Science*, A tenured faculty member engaged in research and instruction. In 2012, founded the *Data, Devices, and Interaction Laboratory* (ddiLab) as a university resource advancing diverse computer science research projects. The lab enabled students to explore how computing can support and extend knowledge and discovery. Played a key role in establishing the department's PhD program and oversaw the graduation of its first doctoral student.
 - Visiting Distinguished Professor (2022 – 2023)
 - PRSA Professor (2020 – 2022)
 - Professor (2017 – 2020)
 - Associate Professor (2012 – 2017)

Illinois Institute of Technology (2021 – 2023)

- **Research Professor**, Department of Computer Science (2021 – 2023), Co-supervised and graduated three PhD students in high performance computing; collaborated with faculty across complementary research areas.

EDUCATION

- **Ph.D. in Computer Science** (2009)
Physical Sciences Division, The University of Chicago, Chicago, IL
Dissertation: *Visualization and Collaboration Technologies to Support High-Performance Computing Research*
Committee: Rick Stevens (Advisor), Todd Dupont, Ian Foster, Mark Hereld, and Ewing Lusk

- **Master of Science in Computer Science** (2002)
Physical Sciences Division, The University of Chicago, Chicago, IL
Thesis: *Development and Analysis of a Movie Playback Tool for Use in Tiled Display Environments*
Advisor: Rick Stevens
- **Master of Science in Electrical Engineering and Computer Science** (1994)
College of Engineering, University of Illinois Chicago, Chicago, IL
Thesis: *Extending Genetic Programming for Discrete Volume Visualization*
Committee: Thomas A. DeFanti (Advisor), Tom Moher, and Daniel J. Sandin
- **Graduate coursework in Physics** (1990 – 1991)
College of Liberal Arts and Sciences, Northern Illinois University, DeKalb, IL
- **Bachelor of Science in Physics** (1990)
College of Liberal Arts and Sciences, Northern Illinois University, DeKalb, IL

Executive Education

- **Alumnus** (2013) Harvard Business School, Harvard University, Cambridge, MA
General Management Program
- **Participant** (2012) The University of Chicago Booth School of Business, Chicago, IL
Essentials of Effective Management: The Psychology of Management
- **Participant** (2012) Hasso Plattner Institute of Design, Stanford University, Stanford, CA
Design Thinking Boot Camp: From Insights to Innovation
- **Participant** (2008 Cohort) The University of Chicago Booth School of Business, Chicago, IL
Argonne's Strategic Laboratory Leadership Program where coursework included: Effective Leadership, Strategic Thinking, Building and Implementing Growth Strategies, and Leading Change and Innovation

Teaching Experience

- **University of Illinois Chicago**
 - Creative Coding (CS 427) (Sp2025)
 - Introduction to High Performance Computing (CS 455) (Sp2023, Sp2024, Sp2025)
 - Honors College Mentoring - (2023 – Present) [3 honors students]
 - Undergraduate Mentoring - (2022 – Present) [Early Research Scholars Team (2023-24)]
- **Northern Illinois University**
 - Big Ideas Lecture (CSCI 600) (Fa2018, Fa2019, Sp2020, Fa2020, Sp2021, Fa2021, Sp2022)
 - Data Structures and Algorithms (CSCI 340) (Sp2014, Sp2017, Sp2020, Fa2020, Sp2021, Fa2021)
 - Data Visualization (CSCI 490, 627) (Fa2012, Fa2013, Sp2015, Sp2016, Sp2019, Sp2022)
 - Information Visualization (CSCI 628) (Sp2020)
 - Introduction to Virtual Reality (CSCI 490, 680) (Sp2018)
 - Independent Study - iOS Graphics (Su2014) [1 student], Data Visualization (Su2016) [7 students], Data Visualization (Fa2016) [12 students], Game Development (Fa2016) [5 students], Data Visualization (Su2017) [8 students], Virtual Reality (Fa2018) [6 students], Virtual Reality (Fa2020) [1 student], Edge Computing (Sp2021) [1 student], Edge Computing (Fa2021) [1 student]

- Undergraduate Mentoring - (2014–2022) [8 Research Rookies, 3 Simmon Scholars, 1 McKearn Fellow, 1 summer research opportunity program participant and 8 senior capstone projects]

AWARDS AND HONORS

- **Best Paper** – *Machine Learning Driven Auto-tuning for Non-uniform All-to-All Collectives*, IEEE International Conference on High Performance Computing, Data, Analytics, and Data Science (HiPC) 2025
- **Best Visualization Award**, *Real-time Scientific Visualization and Interactive Steering for High-Performance Computing Simulations*, PEARC Conference, 2025
- **Distinguished Fellow**, Argonne National Laboratory, 2025
- **Distinguished Senior Research and Development Award**, IEEE Chicago Chapter, 2024
- **Best Paper** – *Bridging Gaps in Simulation Analysis through a General Purpose, Bidirectional Steering Interface with Ascent*, 2024 IEEE/ACM International Workshop on In Situ Infrastructures for Enabling Extreme-Scale Analysis and Visualization (ISAV)
- **Gordon Bell Finalist**, *MProt-DPO: Breaking the ExaFLOPS Barrier for Multimodal Protein Design Workflows with Direct Preference Optimization*, 2024
- **Distinguished Alumni Award**, College of Liberal Arts and Sciences, Northern Illinois University, 2024
- **Best Paper**, *Scaling Computational Fluid Dynamics: In Situ Visualization of NekRS using SENSEI*, 2023 IEEE/ACM International Workshop on In Situ Infrastructures for Enabling Extreme-Scale Analysis and Visualization (ISAV)
- **IEEE SciVis Contest Winner: Best Overall**, *Neuronal Network Simulations of the Human Brain*, 2023
- **IEEE SciVis Contest Winner: Best Workflow**, *Neuronal Network Simulations of the Human Brain*, 2023
- **Argonne National Laboratory Commercialization Excellence Award for Delivering Impact**, 2023
- **Argonne Board of Governors' Distinguished Performance Award**, 2023
- **Honorable Mention, Best Paper** – *ChemoGraph: Interactive Visual Exploration of the Chemical Space*, EuroVis'23
- **ACM Gordon Bell Special Prize for HPC-Based COVID-19 Research** – *GenSLMs: Genome-scale Language Models Reveal SARS-CoV-2 Evolutionary Dynamics*, Supercomputing 2022
- **Argonne Board of Governors' Pinnacle of Education Award**, 2022
- **Best Paper (Nominee)**, **IEEE Transactions on Visualization and Computer Graphics** – *Real-Time Omnidirectional Stereo Rendering: Generating 360° Surround-View Panoramic Images for Comfortable Immersive Viewing*, IEEE VR 2021 (invited high-impact paper presented at SIGGRAPH 2021)
- **The Secretary of Energy Achievement Award**, High Performance Computing Resource Team, 2021
- **Excellence in Teaching Award**, Northern Illinois University Department of Computer Science, 2019–2020

- **Top Recognition for Exemplary Blend of Networking, Computing, and Storage**, *Real-Time Analysis of Streaming Synchrotron Data*, SCinet Technology Challenge, 2019
- **James M. Lufkin Award for Best IPCC Paper**, *Improving Models of Document Cycling: Accounting for the Less Visible Writing Activities of an Annual Reporting Process at a Supercomputing Facility*, IEEE IPCC 2016
- **Honorable Mention, Best Paper**, *Parallel Distributed, GPU-Accelerated, Advanced Lighting Calculations for Large-Scale Volume Visualization*, 6th IEEE Symposium on Large-Scale Data Analysis and Visualization (LDAV 2016)
- **Excellence in Teaching Award**, Northern Illinois University Department of Computer Science, 2015–2016
- **Best Poster**, *Streaming Ultra High Resolution Images to Large Tiled Display at Nearly Interactive Frame Rates with vI3*, 5th IEEE Symposium on Large-Scale Data Analysis and Visualization (LDAV 2015)
- **Best Paper**, *Exploring Void Search for Fault Detection on Extreme Scale Systems*, IEEE Cluster 2014
- **University of Illinois Chicago City Partner Award**, University of Illinois Alumni Association, 2014
- **Star Leadership Award**, Grace McWayne Elementary, 2014 (*for establishing an after-school technology program for fourth and fifth graders*)
- **Argonne National Laboratory Pacesetter**, *Extraordinary Contribution*, 2012
- **ACM Gordon Bell Prize**, *A New Computational Paradigm in Multiscale Simulations: Applications to Blood Flow*, [Honorable Mention], 2011
- **R&D 100 Award**, *Access Grid 3.0*, 2007
- **SC05 HPC Analytics Challenge**, *Real-Time Change Detection and Alerts from Highway Traffic Data*, 2005 International Conference on High Performance Computing, Networking, Storage and Analysis (SC'05)
- **Best Panel**, *Commodity Graphics Accelerators for Scientific Visualization*, 12th Annual IEEE Visualization Conference (VIS 2001)

PUBLICATIONS

Journals, Books, and Book Chapters [Peer Reviewed]

54. R. Koonchanok, M. E. Papka, K. Reda, *Trust Your Gut: Comparing Human and Machine Inference from Noisy Visualizations*, **IEEE Transactions on Visualization and Computer Graphics**, 31(1), pp. 754-764, January 2025, doi:10.1109/TVCG.2024.3456182.
53. A. E. Johnson, L. Renambot, G. E. Marai, D. Tsoupikova, M. E. Papka, L. Long, D. Plepys, J. Talandis, M. D. Brown, J. Leigh, D. J. Sandin, and T. A. DeFanti, *Electronic Visualization Laboratory's 50th Anniversary Retrospective: Look to the Future, Build on the Past*, **PRESENCE: Virtual and Augmented Reality**, 2024. doi.org/10.1162/pres_a_00421
52. 🏆 T. Marrinan, V. A. Mateevitsi, M. Moeller, A. Kanayinkal and M. E. Papka, *VisAnywhere: Developing Multiplatform Scientific Visualization Applications*, **IEEE Computer Graphics and Applications**, 44(5), pp. 93 –103, September-October 2024, doi.org/10.1109/MCG.2024.3444460. [**2023 IEEE Scientific Visualization Contest Winner**]

51. 🏆 M. Zvyagin, A. Brace, K. Hippe, Y. Deng, B. Zhang, C. Orozco Bohorquez, A. Clyde, B. Kale, D. Perez-Rivera, H. Ma, C. M. Mann, M. Irvin, D. G. Ozgulbas, N. Vassilieva, J. G. Pauloski, L. Ward, V. Hayot-Sasson, M. Emani, S. Foreman, Z. Xie, D. Lin and M. Shukla and Weili Nie and J. Romero, C. Dallago, A. Vahdat, C. Xiao, T. Gibbs, I. Foster, J. J. Davis, M. E. Papka, T. Brettin, R. Stevens, A. Anandkumar, V. Vishwanath, A. Ramanathan, *GenSLMs: Genome-scale Language Models Reveal SARS-CoV-2 Evolutionary Dynamics*, **The International Journal of High Performance Computing Applications**, 37(6), pp. 683 – 705, 2023, doi.org/10.1177/10943420231201154. [ACM Gordon Bell Prize]
50. R. Vescovi, R. Chard, N. D. Saint, B. Blaiszik, J. Pruyne, T. Bicer, A. Lavens, Z. Liu, M. E. Papka and S. Narayanan, N. Schwarz, K. Chard, I. T. Foster, *Linking Scientific Instruments and Computation: Patterns, Technologies, and Experiences*, **Patterns**, 3(10), 2022, doi.org/10.1016/j.patter.2022.100606.
49. Y. Fan, B. Li, D. Favorite, N. Singh, T. Childers, P. Rich, W. Allcock, M. E. Papka and Z. Lan, *DRAS: Deep Reinforcement Learning for Cluster Scheduling in High Performance Computing*, **IEEE Transactions on Parallel and Distributed Systems**, 33(12), pp. 4903-4917, 2022, doi.org/10.1109/TPDS.2022.3205325.
48. C. Catlett, P. Beckman, N. Ferrier, M. E. Papka, R. Sankaran, J. Solin, V. Taylor, D. Pancoast, D. Reed, *Hands-On Computer Science: The Array of Things Experimental Urban Instrument*, **Computing in Science and Engineering**, 24(1), pp. 57-63, 2022, doi.org/10.1109/MCSE.2021.3139405.
47. 🏆 T. Marrinan and M. E. Papka, *Real-Time Omnidirectional Stereo Rendering: Generating 360° Surround-View Panoramic Images for Comfortable Immersive Viewing*, **TVCG Special Issue on the 2021 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)**, 2021. [Best Paper Award, Nominee]
46. T. Mackoy, B. Kale, M. E. Papka, R. A. Wheeler, *viewSq, a Visual Molecular Dynamics (VMD) module for calculating, analyzing, and visualizing X-ray and neutron structure factors from atomistic simulations*, **Computer Physics Communications**, 264, 2021, doi.org/10.1016/j.cpc.2021.107881.
45. C. Catlett, P. Beckman; N. Ferrier; H. Nusbaum; M. E. Papka; M. G. Berman; R. Sankaran, *Measuring Cities with Software-Defined Sensors*, **Journal of Social Computing**, 1(1), pp. 14-27, September 2020. doi:10.23919/JSC.2020.0003.
44. T. Marrinan, G. Eisenhauer, M. Wolf, J. A. Insley, S. Rizzi, and M. E. Papka, *Parallel Streaming between Heterogeneous HPC Resources for Real-Time Analysis*, **Journal of Computational Science**, 31, pp. 163-171, February 2019, doi:10.1016/j.jocs.2019.01.003.
43. M. R. Fahey, Y. Alexeev, W. Allcock, B. S. Allen, R. Balakrishnan, A. Benali, L. Booker, A. Boyle, L. Briggs, E. Brooks, P. Carns, B. Cerny, A. Cherry, L. Childers, S. Chunduri, R. Coffey, J. Collins, P. Coffman, S. Coghlan, K. DiBennardi, G. Doyle, H. Finkel, G. Fletcher, M. Garcia, I. Goldberg, C. Goletz, S. Gregurich, K. Harms, C. Holohan, J. A. Insley, T. Jackson, J. Jaseckas, E. Jennings, D. Jensen, W. Jiang, M. Kaczmariski, C. Knight, J. Knowles, K. Kumaran, T. Leggett, B. Lenard, A. Liu, R. Loy, P. Malakar, A. Mantrala, D. E. Martin, G. Mayorga, G. McPheeters, P. Messina, R. Milner, V. Morozov, Z. Nault, D. Nelson, J. O'Connell, J. Osborn, M. E. Papka, S. Parker, P. Patel, S. Patel, E. Pershey, R. Plzak, A. Pope, J. Punzel, S. Ramprakash, J. Reddy, P. Rich, K. Riley, S. Rizzi, G. Rojas, N. A. Romero, R. Scott, A. Scovel, W. Scullin, E. Shemon, H. Siddabathuni Som, J. Stover, M. Suliba, B. Toonen, T. Uram, A. Vazquez-Mayagoitia, V. Vishwanath, R. D. Waldron, G. West, T. J. Williams, D. Wills, L. Wolf, W. Woods, and M. Zhang, *Theta and Mira at Argonne National Laboratory*, **Contemporary High Performance Computing: From Petascale toward Exascale**, J. S. Vetter, Ed., Chapter 2, Volume 3, CRC Press, 2018.

42. L. Yu, Z. Zhou, Y. Fan, M. E. Papka, and Z. Lan, *System-wide trade-off modeling of performance, power, and resilience on petascale systems*, **The Journal of Supercomputing**, 74, pp. 3168–3192, 2018, doi.org/10.1007/s11227-018-2368-8.
41. M. Kostuk, T. D. Uram, T. Evans, D. M. Orlov, M. E. Papka, D. Schissel, *Automatic Between-Pulse Analysis of DIII-D Experimental Data Performed Remotely on a Supercomputer at Argonne Leadership Computing Facility*, **Fusion Science and Technology**, 2(1), pp. 1-9, 2018, doi:10.1080/-15361055.2017.1390388.
40. J. T. Childers, T. D. Uram, T. J. LeCompte, M. E. Papka, D. P. Benjamin, *Adapting the Serial Alpgen Parton-Interaction Generator to Simulate LHC Collisions on Millions of Parallel Threads*, **Computer Physics Communications** 210, pp. 54-59, 2016.
39. K. Reda, A. E. Johnson, M. E. Papka, J. Leigh, *Modeling and Evaluating User Behavior in Exploratory Visual Analysis*, **Information Visualization** 15(4), pp. 325-339, October 2016.
38. W. Usher, I. Wald, A. Knoll, M. Papka, V. Pascucci, *In Situ Exploration of Particle Simulations with CPU Ray Tracing*, **Supercomputing Frontiers and Innovations** 3(4), pp. 4-18, October 2016.
37. S. Wallace, Z. Zhou, V. Vishwanath, S. Coghlan, J. Tramm, Z. Lan, M. E. Papka, *Application Power Profiling on IBM Blue Gene/Q*, **Parallel Computing** 57, pp. 73-86, September 2016.
36. P. Perdikaris, J.A. Insley, L. Grinberg, Y. Yu, M. E. Papka, G. E. Karniadakis, *Visualizing Multiphysics, Fluid-Structure Interaction Phenomena in Intracranial Aneurysms*, **Parallel Computing** 55, pp. 9-16, July 2016.
35. A. Gyulassy, A. Knoll, K. C. Lau, B. Wang, P.-T. Bremer, M. E. Papka, L. Curtiss, V. Pascucci, *Interstitial and Interlayer Ion Diffusion Geometry Extraction in Graphitic Nanosphere Battery Materials*, **IEEE Transactions on Visualization and Computer Graphics** 22(1):916-925, January 2016.
34. H. Bui, E.-S. Jung, V. Vishwanath, A. Johnson, J. Leigh, M. E. Papka, *Improving Sparse Data Movement Performance using Multiple Paths on the Blue Gene/Q Supercomputer*, **Parallel Computing** 51, pp. 3-16, January 2016.
33. R. Chard, R. Madduri, N. T. Karonis, K. Chard, K. Duffin, C. E. Ordonez, T. D. Uram, J. Fleischauer, I. T. Foster, M. E. Papka, J. Winans, *Scalable pCT Image Reconstruction Delivered as a Cloud Service*, **IEEE Transactions on Cloud Computing**, 2015, doi.org/10.1109/TCC.2015.2457423.
32. L. Yu, Z. Zhou, S. Wallace, M. E. Papka, Z. Lan, *Quantitative Modeling of Power Performance Tradeoffs on Extreme Scale Systems*, **Journal of Parallel and Distributed Computing**, 84(C), pp. 1-14, 2015, doi.org/10.1016/j.jpdc.2015.06.006.
31. V. Vishwanath, H. Bui, M. Hereld, M. E. Papka, *GLEAN*. In **High Performance Parallel I/O**, Prabhat and Quincey Koziol, Eds., Boca Raton: Chapman and Hall/CRC Computational Science, pp. 215-223, 2014.
30. A. Knoll, I. Wald, P. Navratil, A. Bowen, K. Reda, M. E. Papka, K. Gaither, *RBF Volume Ray Casting on Multicore and Manycore CPUs*, **Computer Graphics Forum**, 33(3):71-80, 2014.
29. S. Lakshminarasimhan, X. Zou, D. A. Boyuka II, S. V. Pendse, J. Jenkins, V. Vishwanath, M. E. Papka, S. Klasky, N. F. Samatova, *DIRAQ: Scalable In Situ Data- and Resource-Aware Indexing for Optimized Query Performance*, **Cluster Computing**, 17(63):1-19, 2014.
28. N. T. Karonis, K. L. Duffin, C. E. Ordonez, B. Erdelyi, T. D. Uram, E. C. Olson, G. Coutrakon, M. E. Papka, *Distributed and Hardware Accelerated Computing for Clinical Medical Imaging Using Proton Computed Tomography (pCT)*, **Journal of Parallel and Distributed Computing**, 73(12):1605-1612, 2013.

27. A. Knoll, M. KY Chan, K. C. Lau, B. Liu, J. Greeley, L. Curtiss, M. Hereld, M. E. Papka, *Uncertainty Classification and Visualization of Molecular Interfaces*, **International Journal for Uncertainty Quantification**, 3(2):157-69, 2013.
26. K. Reda, A. Febretti, A. Knoll, J. Aurisano, J. Leigh, A. E. Johnson, M. E. Papka, M. Hereld, *Visualizing Large, Heterogeneous Data in Hybrid-Reality Environments*, **IEEE Computer Graphics and Applications**, 33(4):38-48, 2013.
25. W. E. Allcock, A. M. Bailey, R. Bair, C. Bacon, R. Balakrishnan, A. Bertsch, B. Bihari, B. Carnes, D. Chen, G. Chiu, R. Coffey, S. Coghlan, P. Coteus, K. Cupps, E. W. Draeger, T. W. Fox, L. Fried, M. Gary, J. Glosli, T. Gooding, J. Gunnels, J. Gyllenhaal, J. Hammond, R. Haring, P. Heidelberger, M. Hereld, T. Inglett, K.H. Kim, K. Kumaran, S. Langer, A. Mamidala, R. McCallen, P. Messina, S. Miller, A. Mirin, V. Morozov, F. Najjar, M. Nelson, A. Nichols, M. Ohmacht, M. E. Papka, F. Petrini, T. Quinn, D. Richards, N. A. Romero, K. D. Ryu, A. Schram, R. Shearer, T. Spelce, B. Springmeyer, F. Streitz, B. de Supinski, P. Vranas, B. Walkup, A. Wang, T. Williams, and R. Wisniewski, *Blue Gene/Q: Sequoia and Mira*, In **Contemporary High Performance Computing: From Petascale toward Exascale**, J. S. Vetter, Ed., Boca Raton: Chapman and Hall/CRC Computational Science, 2013, pp. 225-281.
24. G. Bebis, R. Boyle, B. Parvin, D. Koracin, F. Charless, W. Sen, C. Min-Hyung, S. Mantler, J. Schulze, D. Acevedo, K. Mueller, M. E. Papka, Eds., *Advances in Visual Computing - 8th International Symposium, ISVC 2012, Rethymnon, Crete, Greece, July 16-18, 2012, Revised Selected Papers, Part II.*, **Lecture Notes in Computer Science 7432**, Springer 2012.
23. G. Bebis, R. Boyle, B. Parvin, D. Koracin, F. Charless, W. Sen, C. Min-Hyung, S. Mantler, J. Schulze, D. Acevedo, K. Mueller, M. E. Papka, Eds., *Advances in Visual Computing - 8th International Symposium, ISVC 2012, Rethymnon, Crete, Greece, July 16-18, 2012, Revised Selected Papers, Part I.*, **Lecture Notes in Computer Science 7431**, Springer 2012.
22. L. Grinberg, J. A. Insley, D. Fedosov, V. A. Morozov, M. E. Papka, G. E. Karniadakis, *Tightly Coupled Atomistic-Continuum Simulations of Brain Blood Flow on Petaflop Supercomputers*, **IEEE Computing in Science and Engineering**, 14(6):58-67, 2012.
21. J. Dongarra, P. Beckman, T. Moore, P. Aerts, Gi Aloisio, J-C. Andre, D. Barkai, J-Y. Berthou, T. Boku, B. Braunschweig, F. Cappello, B. Chapman, X. Chi, A. Choudhary, S. Dosanjh, T. Dunning, S. Fiore, A. Geist, B. Gropp, R. Harrison, M. Hereld, M. Heroux, A. Hoisie, K. Hotta, Z. Jin, Y. Ishikawa, F. Johnson, S. Kale, R. Kenway, D. Keyes, B. Kramer, J. Labarta, A. Lichnewsky, T. Lippert, B. Lucas, B. Maccabe, S. Matsuoka, P. Messina, P. Michielse, B. Mohr, M. S. Mueller, W. E. Nagel, H. Nakashima, M. E. Papka, D. Reed, M. Sato, E. Seidel, J. Shalf, D. Skinner, M. Snir, T. Sterling, R. Stevens, F. Streitz, B. Sugar, S. Sumimoto, W. Tang, J. Taylor, R. Thakur, A. Trefethen, M. Valero, A. van der Steen, J. Vetter, P. Williams, R. Wisniewski and K. Yelick, *The International Exascale Software Project Roadmap*, **International Journal of High Performance Computing Applications**, 25(1):3-60, 2011.
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10. T. Disz, M. E. Papka, R. Stevens *UbiWorld: An Environment Integrating Virtual Reality, Supercomputing, and Design*, **Proceedings of the 6th Heterogeneous Computing Workshop (HCW'97)**, pp. 46-57, Geneva, Switzerland, April 1, 1997.
9. I. Foster, M. E. Papka, R. Stevens, *Tools for Distributed Collaborative Environments: A Research Agenda*, **Proceedings of the 5th IEEE International Symposium on High Performance Distributed Computing**, pp. 23-8, Syracuse, NY, August 6-9, 1996.
8. M. E. Papka, R. Stevens *UbiWorld: An Environment Integrating Virtual Reality, Supercomputing, and Design*, **Proceedings of the 5th IEEE International Symposium on High Performance Distributed Computing**, pp. 306-7, Syracuse, NY, August 6-9, 1996.
7. T. Canfield, T. L. Disz, M. E. Papka, R. Stevens, M. Huang, V. Taylor, J. Chen, *Toward Real-Time Interactive Virtual Prototyping of Mechanical Systems: Experiences Coupling Virtual Reality with Finite Element Analysis*, **Proceedings of High Performance Computing '96**, New Orleans, LA, March 31-April 1, 1996.
6. T. L. Disz, M. E. Papka, M. Pellegrino, R. Stevens, *Sharing Visualization Experiences among Remote Virtual Environments*, **Proceedings of the International Workshop on High Performance Computing for Computer Graphics and Visualisation**, pp. 217-37, Swansea, Wales, July 3-4, 1995.
5. T. Disz, M. E. Papka, R. Stevens, M. Pellegrino, V. Taylor, *Virtual Reality Visualization of Parallel Molecular Dynamics Simulation*, **1995 Simulation Multiconference Symposium**, pp. 483-87, Phoenix, AZ, April 1995.
4. L. R. Turner, D. Levine, M. Huang, M. E. Papka, L. Kettunen, *Using the CAVE Virtual-Reality Environment as an Aid to 3-D Electromagnetic Field Computation*, **Proceedings of the Conference on the Computation of Electromagnetic Fields (COMPUMAG-Berlin)**, Berlin, Germany, July 10-13, 1995.

3. M. Huang, M. E. Papka, T. DeFanti, D. Levine, L. Turnery, L. Kettunen, *Virtual Reality Visualization of Accelerator Magnets*, **Proceedings of the SCS Simulation MultiConference: Simulation as a Critical Technology**, Phoenix, AZ, April 9-13, 1995.
2. S. Das, T. Franguiadakis, M. E. Papka, T. A. DeFanti, D. J. Sandin, *A Genetic Programming Application in Virtual Reality*, **Proceedings of the First IEEE Conference on Evolutionary Computation, IEEE World Congress on Computational Intelligence**, pp. 480-84, Orlando, FL, June 27-29, 1994.
1. C. Cruz-Neira, J. Leigh, M. E. Papka, C. Barnes, S. M. Cohen, S. Das, R. Engelmann, R. Hudson, T. Roy, L. Siegel, *Scientists in Wonderland: A Report on Visualization Applications in the CAVE Virtual Reality Environment*, **Proceedings of the IEEE 1993 Symposium on Research Frontiers in Virtual Reality**, pp. 59-66, San Jose, CA, October 25-26, 1993.

Technical Reports and Other Publications [Non-Peer Reviewed]

36. L. C. McInnes, D. Arnold, P. Balaprakash, M. Bernhardt, B. Cerny, A. Dubey, R. Giles, D. W. Hood, M. A. Leung, V. Lopez-Marrero, P. Messina, O. B. Newton, C. Oehmen, S. M. Wild, J. Willenbring, L. Woodley, T. Baylis, D. E. Bernholdt, C. Camano, J. Cohoon, C. Ferenbaugh, S. M. Fiore, S. Gesing, D. Gomez-Zara, J. Howison, T. Islam, D. Kepczynski, C. Lively, H. Menon, B. Messer, M. Ngom, U. Paliath, M. E. Papka, I. Qualters, E. M. Raybourn, K. Riley, P. Rodriguez, D. Rouson, M. Schwalbe, S. K. Seal, O. Surer, V. Taylor, and L. Wu. *Report of the 2025 Workshop on Next-Generation Ecosystems for Scientific Computing: Harnessing Community, Software, and AI for Cross-Disciplinary Team Science*, **arXiv:2510.03413**, 2025.
- Z. Zhu, Y. Sun, D. Parakal, B. Fang, S. Farrell, G. H. Bauer, B. Bode, I. T. Foster, M. E. Papka, W. Gropp, Z. Zhang, and L. Yang, *Understanding the Landscape of Ampere GPU Memory Errors*, **arXiv:2508.03513**, 2025.
35. Z. Zheng, S. Sultanov, M. E. Papka, Z. Lan, *Exploring Uncore Frequency Scaling for Heterogeneous Computing*, **arXiv:2502.03796**, 2025.
34. Shilpika, B. Lusch, V. Vishwanath, M. E. Papka, *An Incremental Multi-Level, Multi-Scale Approach to Assessment of Multifidelity HPC Systems*, **arXiv:2501.17796v1**, 2025.
33. Z. Zheng, M. E. Papka, Z. Lan, *More for Less: Integrating Capability-Predominant and Capacity-Predominant Computing*, **arXiv:2501.12464v1**, 2025.
32. X. Ma, F. Yan, L. Yang, I. Foster, M. E. Papka, Z. Liu, R. Kettimuthu, *MalleTrain: Deep Neural Network Training on Unfillable Supercomputer Nodes*, **arXiv:2404.15668v1**, 2024.
31. B. Li, Y. Fan, M. Dearing, Z. Lan, P. Rich, W. Allcock, M. E. Papka, *MRSch: Multi-Resource Scheduling for HPC*, **arXiv:2403.16298v2**, 2024.
30. S. L. Song, B. Krufft, M. Zhang, C. Li, S. Chen, C. Zhang, M. Tanaka, X. Wu, J. Rasley, A. A. Awan, C. Holmes, M. Cai, A. Ghanem, Z. Zhou, Y. He, P. Luferenko, D. Kumar, J. Weyn, R. Zhang, S. Klocek, V. Vragov, M. AlQuraishi, G. Ahdritz, C. Floristean, C. Negri, R. Kotamarthi, V. Vishwanath, A. Ramanathan, S. Foreman, K. Hippe, T. Arcomano, R. Maulik, M. Zvyagin, A. Brace, B. Zhang, C. Orozco Bohorquez, A. Clyde, B. Kale, D. Perez-Rivera, H. Ma, C. M. Mann, M. Irvin, J. G. Pauloski, L. Ward, V. Hayot, M. Emani, Z. Xie, D. Lin, M. Shukla, I. Foster, J. J. Davis, M. E. Papka, T. Brettin, P. Balaprakash, G. Tourassi, J. Gounley, H. Hanson, T. E. Potok, M. L. Pasini, K. Evans, D. Lu, D. Lunga, J. Yin, S. Dash, F. Wang, M. Shankar, I. Lyngaas, X. Wang, G. Cong, P. Zhang, M. Fan, S. Liu, A. Hoisie, S. Yoo, Y. Ren, W. Tang, K. Felker, A. Svyatkovskiy, H. Liu, A. Aji, A. Dalton, M. Schulte, K. Schulz, Y. Deng, W. Nie, J. Romero, C. Dallago, A. Vahdat, C. Xiao, T. Gibbs, A. Anandkumar, R. Stevens, *DeepSpeed4Science Initiative: Enabling Large-Scale Scientific Discovery through Sophisticated AI System Technologies*, **arXiv:2310.04610**, 2023.

29. Y. Fan, P. Rich, W. Allcock, M. E. Papka, Z. Lan, *Hybrid Workload Scheduling on HPC Systems*, **arXiv:2109.05412**, 2021.
28. Z. Liu, R. Kettimuthu, M. E. Papka, I. Foster, *BFTrainer: Low-Cost Training of Neural Networks on Unfillable Supercomputer Nodes*, **arXiv:2106.12091**, 2021.
27. M. Salim, T. Uram, J. T. Childers, V. Vishwanath, M. E. Papka, *Toward Real-time Analysis of Experimental Science Workloads on Geographically Distributed Supercomputers*, **arXiv:2105.06571**, 2021.
26. Y. Fan, Z. Lan, J. T. Childers, P. Rich, W. Allcock, M. E. Papka, *Deep Reinforcement Agent for Scheduling in HPC*, **arXiv:2102.06243**, 2021.
25. M. Emani, V. Vishwanath, C. Adams, M. E. Papka, R. Stevens, L. Florescu, S. Jairath, W. Liu, T. Nama, A. Sujeeth, *Accelerating Scientific Applications with SambaNova Reconfigurable Dataflow Architecture*, **Computing in Science & Engineering**, 2021.
24. J. J. Hack and M. E. Papka *The US High-Performance Computing Consortium in the Fight Against COVID-19*, **IEEE Computing in Science and Engineering**, 22(6), pp. 75-80, October 2020.
23. S. Collis, P. Beckman, E. Kelly, C. Catlett, R. Sankaran, I. Altintas, J. Olds, N. Ferrier, S. Park, Y. Kim, M. E. Papka. *Introducing Sage: Cyberinfrastructure for Sensing at the Edge*, **EGU General Assembly Conference Abstracts**, May 2020.
22. M. A. Salim, T. D. Uram, J. T. Childers, P. Balaprakash, V. Vishwanath, M. E. Papka, *Balsam: Automated scheduling and execution of dynamic, data-intensive HPC workflows*, **arXiv:1909.08704**, 2019.
21. R. Gerber, J. Hack, K. Riley, K. Antypas, R. Coffey, E. Dart, T. Straatsma, J. Wells, D. Bard, S. Dosanjh, I. Monga, M. E. Papka, L. Rotman. *Exascale Requirements Reviews, March 9-10, 2017 – Tysons Corner, Virginia: Crosscut Report from an Office of Science review sponsored by: Advanced Scientific Computing Research, Basic Energy Sciences, Biological and Environmental Research, Fusion Energy Sciences, High Energy Physics, Nuclear Physics*, January 2018.
20. J. Vetter, A. Almgren, P. DeMar, K. Riley, K. Antypas, D. Bard, R. Coffey, E. Dart, S. Dosanjh, R. Gerber, J. Hack, I. Monga, M. E. Papka, L. Rotman, T. Straatsma, J. Wells, D. E. Bernholdt, W. Bethel, G. Bosilca, F. Cappello, T. Gamblin, S. Habib, J. Hill, J. K. Hollingsworth, L. Curfman McInnes, K. Mohror, S. Moore, K. Moreland, R. Roser, S. Shende, G. Shipman, S. Williams. *Advanced Scientific Computing Research Exascale Requirements Review, September 27-29, 2016 – Rockville, Maryland: Report from an Office of Science review sponsored by Advanced Scientific Computing Research*, 2017.
19. M. E. Papka, P. Messina, L. Wolf, B. Cerny, J. Collins. *Argonne Leadership Computing Facility 2016 Annual Report. Technical Report ANL-01/17*, Argonne National Laboratory, Argonne, Illinois.
18. T. D. Uram, M. E. Papka, *Expanding the Scope of High-Performance Computing Facilities*, **IEEE/AIP Computing in Science and Engineering**, May/June, 2016.
17. M. E. Papka, P. Messina, L. Wolf, B. Cerny, J. Collins. *Argonne Leadership Computing Facility 2015 Annual Report. Technical Report ANL-01/16*, Argonne National Laboratory, Argonne, Illinois.
16. J. J. Hack, M. E. Papka, *Big Data: Next-Generation Machines for Big Science*, **IEEE/AIP Computing in Science and Engineering**, 17(4), pp. 63-65, July-Aug. 2015.
15. M. E. Papka, P. Messina, L. Wolf, B. Cerny, J. Collins. *Argonne Leadership Computing Facility 2014 Annual Report. Technical Report ANL-01/15*, Argonne National Laboratory, Argonne, Illinois.
14. M. E. Papka, P. Messina, L. Wolf, B. Cerny, J. Collins. *Argonne Leadership Computing Facility 2013 Annual Report. Technical Report ANL-04/14*, Argonne National Laboratory, Argonne, Illinois.

13. M. E. Papka, P. Messina, L. Wolf, B. Cerny, J. Collins. *Argonne Leadership Computing Facility 2012 Annual Report*. **Technical Report ANL-06/13**, Argonne National Laboratory, Argonne, Illinois.
12. M. E. Papka, P. Messina, R. Coffey, C. Drugan. *Argonne Leadership Computing Facility 2011 Annual Report: Shaping Future Supercomputing*. **Technical Report ANL-12/22**, Argonne National Laboratory, Argonne, Illinois.
11. M. E. Papka, P. Messina, R. Coffey, C. Drugan. *Argonne Leadership Computing Facility 2010 Annual Report: Shaping Future Supercomputing*. **Technical Report ANL-11/5**, Argonne National Laboratory, Argonne, Illinois.
10. S. Ahern, A. Shoshani, K.-L. Ma, A. Choudhary, T. Critchlow, S. Klasky, V. Pascucci, J. Ahrens, W. Bethel, H. Childs, J. Huang, K. Joy, Q. Koziol, G. Lofstead, J. Meredith, K. Moreland, G. Ostrouchov, M. E. Papka, V. Vishwanath, M. Wolf, N. Wright, K. Wu. *Scientific Discovery at the Exascale: Report from the DOE/ASCR 2011 Workshop on Exascale Data Management, Analysis, and Visualization*, 2011.
9. C. Johnson, R. Ross, S. Ahern, J. Ahrens, W. Bethel, K.-L. Ma, M. E. Papka, J. van Rosendale, H.-W. Shen, J. Thomas, *Visualization and Knowledge Discovery: Report from DOE/ASCR Workshop on Visual Analysis and Data Exploration at Extreme Scale*, 2007.
8. D. Schissel, J. Burruss, A. Finkelstein, et al. *SciDAC Fusiongrid Project – A National Collaboration to Advance the Science of High Temperature Plasma Physics for Magnetic Fusion*. **Technical Report DOE/ER25456**, The Trustees of Princeton University; Princeton Plasma Physics Laboratory; Massachusetts Institute of Technology; Lawrence Berkeley National Laboratory; Argonne National Laboratory; University of Utah, August 2006.
7. J. A. Insley and M. E. Papka, *Prototyping Simple Access to Visualization Resources*, **ClusterWorld**, June 2005.
6. N. Karonis, M. E. Papka, J. Binns, J. Bresnahan, and J. Link. *Effective Use of Dedicated Wide-Area Networks for High-Performance Distributed Computing*. **Technical Report ANL/MCS-P1151-0404**, Argonne National Laboratory, Argonne, Illinois, 2004.
5. T. Udeshi, R. Hudson, and M. E. Papka. *Seamless Multiresolution Isosurfaces Using Wavelets*. **Technical Report ANL/MCS-P801-0300**, Argonne National Laboratory, Argonne, Illinois, March 2000.
4. J. Ahrens, C. Law, W. Schroeder, K. Martin, and M. E. Papka. *A Parallel Approach for Efficiently Visualizing Extremely Large, Time-Varying Datasets*. **Technical Report LAUR-00-1620**, Los Alamos National Laboratory, Los Alamos, New Mexico, 2000.
3. J. Bresnahan, J. Insley, and M. E. Papka. *Interacting with Scientific Visualizations: User-Interface Tools within Spatially Immersive Displays*. **Technical Report ANL/MCS-P789-0100**, Argonne National Laboratory, Argonne, Illinois, 2000.
2. T. Disz, R. Olson, M. E. Papka, R. Stevens, M. Szymanski, and R. Firby. *Two Implementations of Shared Virtual Space Environments*. **Technical Report ANL/MCS-P652-0297**, Argonne National Laboratory, Argonne, Illinois, 1997.
1. T. Disz, M. E. Papka, M. Pellegrino, and R. Stevens. *CAVEcomm User Manual*. **Technical Memorandum ANL/MCS-TM-218**, Argonne National Laboratory, Argonne, IL, December 1996.

RESEARCH GRANTS and CONTRACTS

Leadership and engagement in more than \$31M of DOD, DOE, NIH, NSF, and other funded research. Oversight and management of more than \$2B in funding as director (PI) of the ALCF.

Current Grants

11. *Argonne Leadership Computing Facility (ALCF)*, DOE ASCR, \$42M (2010), \$62M (2011), \$62M (2012), \$61M (2013), \$67M (2014), \$80M (2015), \$81M (2016), \$80M (2017), \$110M (2018), \$140M (2019), \$150M (2020), \$150M (2021), \$160M (2022), \$175M (2023), \$219M (2024), \$219M (2025), \$238M (2026) (PI, Argonne), Ongoing
10. *American Science Cloud* DOE ASCR, \$4M, (PI, Argonne), 2025 – 2026
9. *Sage Grande: An Open Artificial Intelligence Testbed for Edge Computing and Intelligent Sensing*, NSF (OAC-2436842), \$1,035,000 (PI, UIC, subcontract Northwestern University, PI: Beckman) 2025 – 2030
8. *SHF: Small: Beyond Performance: Energy-Conscious High Performance Computing*, NSF (CCF-2440948), \$540,000, (co-PI, UIC), 2025 – 2028
7. *NAIRR Software Workshop*, NSF (OAC-2440948), \$99,739, (PI, UIC), 2024 – 2025
6. *AIMCI: AI-Guided Resource Management for Advanced Cyberinfrastructure*, NSF (OAC-2402901), \$600,000, (co-PI, UIC), 2024 – 2027
5. *Digital Twins and AI-enabled Immersive Environments for Automated Scientific Laboratories (DAIMSL)*, DOE ASCR, \$182,999, (PI, UIC, subcontract Argonne National Laboratory, PI: Ferrier), 2024 – 2026
4. *A National-Scale Testbed Supporting Artificial Intelligence Research Spanning the Computing Continuum*, NSF (OAC-2331263), \$120,000, (PI, UIC, subcontract Northwestern University, PI: Beckman), 2023 – 2026
3. *Research Infrastructure: MRI: Track 2 Acquisition of Data Observation and Computation Collaboratory (DOCC)*, NSF (CNS-2320261), \$1,548,545, (co-PI, UIC), 2023 – 2026
2. *Tele-robotics to Tele-autonomous Robotics for Isotope Production*, DOE ASCR/Isotope, \$70,716, (PI, UIC, subcontract Argonne National Laboratory, PI: Nolan), 2023 – 2025
1. *Chicagoland Computational Traineeship in High Energy Particle Physics*, DOE HEP, \$650,000, (co-PI, UIC, subcontract Northern Illinois University, PI: Adelman), 2022 – 2027

Past Grants

24. *Data Sharing and Analysis Across Leadership Computing Facilities* DOE HEP, \$99,686, (PI, UIC, subcontract Argonne National Laboratory, PI: Heitmann), 2023 – 2025
23. *CyberTraining: Implementation: Medium: FOUNT: Scaffolded, Hands-On Learning for a Data-Centric Future*, NSF (OAC-2230080), \$204,815, (co-PI, NIU), 2022 – 2025
22. *NSF Mid-Scale RI-1: SAGE: A Software-Defined Sensor Network*, NSF (OAC-1935984), \$9,026,927 (PI: Beckman@Northwestern, Subcontract to NIU: \$80,000), 2019 – 2022
21. *Collaborative Research: PPOSS: Planning: A Heterogeneous HPC Environment for AI-Enabled Science*, NSF, \$40,000, (co-PI, NIU), 2021 – 2022
20. *Presidential Research, Scholarship, and Artistry Professor*, Northern Illinois University, \$20,000, (PI, NIU), 2020 – 2022

19. *COVID Research and Resources*, DOE, \$7,152,000, (PI, Argonne), 2020 – 2021
18. *MRI: Development of an Urban-Scale Instrument for Interdisciplinary Research*, NSF (OAC-1532133), \$3,514,285 (PI: Catlett@UChicago, NIU portion: \$61,401), (co-PI, NIU), 2015 – 2019
17. *CC*IIIE Integration: Collaborative Research: EPSON: Embracing Parallel Networks and Storage for Predictable End-to-End Data Movement*, NSF (OAC-1440797), \$279,928, (PI, NIU), 2014 – 2017
16. *pCT Imaging Using GPGPUs and Centralized Facilities*, DOD (via NIU), \$1,041,697, (PI, Argonne), 2010 – 2013
15. *Collaborative Research: Scalable Multiscale Models for the Cerebrovasculature: Algorithms, Software and Petaflop Simulations*, NSF (OAC-0904190), \$276,510, (PI, UChicago), 2009 – 2013
14. *MRI: Acquisition of PADS - A Petscale Active Data Store*, NSF (OAC-0821678), \$1,102,135, (PI, UChicago), 2008 – 2012
13. *Collaborative Visualization and the Analysis Pipeline*, DOE, \$650,000, (PI, Argonne), 2007 – 2009
12. *Social Informatics Grid (SIDGrid)*, NSF (BCS-0537849), \$2,029,933, (Senior Personnel, UChicago), 2005 – 2007
11. *SCI: TeraGrid Resource Partners*, NSF (OAC-0504086), \$3,614,244, (co-PI, UChicago), 2007 – 2011
10. *SCI: TeraGrid Early Operations*, NSF (OAC-0451491), \$1,058,536, (co-PI, UChicago), 2005 – 2007
9. *Middleware to Support Group to Group Collaboration*, DOE, \$800,000, (co-PI, Argonne), 2004 – 2006
8. *National Fusion Collaboratory*, DOE, \$440,000, (Senior Personnel, Argonne), 2004 – 2006
7. *Grid Visualization*, DOE (subcontract from Kitware Inc.), \$70,000, (PI, Argonne), 2004 – 2006
6. *Advanced Biomedical Collaboration Testbed in Surgery, Anesthesia, and Emergency Medicine (ABC Testbed)*, NIH, \$3,487,139, (Senior Personnel, UChicago), 2003 – 2006
5. *A Data Intense Challenge: The Instrumented Oilfield of the Future*, NSF (CNS-0121523), \$1,461,005 (PI: Wheeler@UTexas, UChicago portion: \$384,000), (Senior Personnel, UChicago), 2002 – 2005
4. *Extensible Network Services for the Access Grid*, NSF (OAC-0222509), \$900,000, (co-PI, UChicago), 2002 – 2005
3. *Scientific Workspaces of the Future*, NSF (via UIUC), \$800,000, (co-PI, Argonne), 2002 – 2004
2. *Advanced Display Work*, General Atomics Inc., \$90,000, (PI, Argonne), 2002 – 2003
1. *Advanced Visualization Technology Center*, DOE, \$1,750,000, (co-PI, Argonne), 1999 – 2002

SYNERGISTIC ACTIVITIES - LAST 10 YEARS

Professional Memberships

- Member (2024), *Computer Science Teachers Association* (CSTA)
- Distinguished Member (2023), *Association for Computing Machinery* (ACM) – For contributions in virtual reality, collaborative environments, scientific visualization, and high-performance computing research and operations.
- Senior Member (2000), *Institute of Electrical and Electronics Engineers* (IEEE)
- Member (2008), *American Association for the Advancement of Science* (AAAS)

Chair, Steering, and Special Engagements

- IEEE Vis 2027 (to be hosted in Chicago), General Co-Chair
- IEEE VIS Area Curation Committee (2022 – 2025)
- Practice and Experience in Advanced Research Computing (PEARC) (2021), Co-Chair, Visualization Showcase with Joseph Insley (ANL/NIU)
- Symposium on Large Data Analysis and Visualization, Founder (with Jim Ahrens, Chris Johnson, Kwan-Li Ma) and Steering Committee Member (2010 – 2019)
- IEEE Vis 2015 (hosted in Chicago), General Co-Chair with Maxine Brown (UIC)

Program Committees

- International Conference for High Performance Computing, Networking, Storage, and Analysis
 - Technical Program [State of the Practice] - Chair (2025)
 - Technical Program [Art of HPC] - Member (2024)
 - Technical Program [Data Analytics, Visualization & Storage] - Member (2023)
 - Technical Program [Data Analytics, Visualization & Storage] - Member (2018)
- IEEE VIS
 - Workshops Co-Chair (2025)
 - Workshop Reviewer (2025)
 - Technical Committee (2025)
 - Full Paper Reviewer (2016, 2023 – 2025)
 - Short Paper Reviewer (2024)
- EuroVis
 - Reviewer Education Papers (2024)
 - Reviewer Full Papers (2020, 2021, 2023)
- ACM Conference on Intelligent User Interfaces, Reviewer Posters and Demos (2024, 2025)
- Parallel AI and Systems for the Edge (2020 – 2025)
- 21st IEEE International eScience Conference 2025

- 32nd IEEE International Conference on High Performance Computing, Data, and Analytics - Scalable Data Science (Data Science and AI) 2025
- IEEE International Symposium on Mixed and Augmented Reality (ISMAR) 2024

Reviewer

- International Conference for High Performance Computing, Networking, Storage, and Analysis
- International Symposium on Visual Computing
- IEEE Vis Conference (Visualization, Information, and Visualization and Analysis)
- IEEE Transactions on Computers 2026
- National Science Foundation
 - NSF Major Research Instrumentation
 - Strategic Technologies for Cyberinfrastructure Programs
- various other workshops and conferences ...

Advisory Board Memberships

- **Advisory Board Member:**
 - The Ohio State University Computer Science and Engineering Department External Advisory Committee (2025 – Present)
 - Loyola University Chicago Computer Science Advisory Committee (2015 – 2025)
 - Harvard University Computational Science and Engineering Advisory Board (2012 – 2022)
 - NSF-funded Project: Precipitating Change – Integrating Meteorology, Mathematics, and Computational Thinking (2016 – 2019)
- **Associate Editor:** Frontiers in High Performance Computing (*Architecture and Systems*) (2022 – Present)
- **Editorial Board Member:** IEEE Computing in Science and Engineering (CiSE) – *Department Editor for Leadership Computing* (2015 – 2020)
- **Internal Service at Argonne National Laboratory:**
 - Executive Sponsor, Argonne African American Employee Resource Group (2019 – 2025)
- **Internal Service at the University of Illinois Chicago:**
 - CS + Game Curriculum Development (2025 – Present) - Member
 - Provost Search Committee (2025 – 2026) - Member
 - Computer Science Advisory Committee (2025 – 2026) - Member
 - Computer Science Computing Committee (2022 – Present) – Member; (2024 – 2026) - Chair
 - Computer Science Promotion and Tenure Committee (2025 – 2026) - Member
 - Computer Science Department Head Search Committee (2024 – 2025) – Member
 - Dean of Engineering Search Committee (2023 – 2024) – Member
- **University of Illinois System Research Computing Advisory Committee** (2024 – Present) – Member

Advisees Graduation Dates in []

- **Current Postdocs:**

- Yuping Fan (ANL)
- Shilpika (ANL)

- **Current Graduate Students:**

- Idunnuoluwa Adeniji [Ph.D. Student]
- Mahsa Alishiri [MS Student]
- Niccolo Brembilla [MS 2025, PhD Student]
- Hal Brynteson [M.S. Student]
- Andrew Burks [Ph.D. Student]
- Amy Byrnes [M.S. 2024, Ph.D. Student] C² the P² Fellow
- Greg Cross [ECE Ph.D. Student]
- Smita Darmora [MS Student]
- Chris Grams [M.S. 2024, Ph.D. Student] [co-advisor with Yu (Tom) Gao (PHAR)]
- Hemalatha Ningappa Kondakundi [M.S. Student]
- Shreyas Kulkarni [Ph.D. Student] [co-advisor with Saeed Boorboor (CS)]
- Yash Kurkure [M.S. 2023, Ph.D. Student]
- Wesley Kwiecinski [M.S. Student] C² the P² Fellow
- Kim Meagher [M.S. Student] C² the P² Fellow
- Jyotsna Rajaraman [Ph.D. Student]
- Aaqel Shaik [BS 2025, MS Student]
- Zhong Zheng [Ph.D. Student] [co-advisor with Zhiling Lan (CS)]

- **Past Graduate Students:**

- Carla Floricel [Ph.D. 2025] (UIC) [Advisor]
- Marco Bonafini [M.S. 2024] (UIC) [co-advisor with Ian Kash (CS)]
- Riccardo Strina [M.S. 2024] (UIC) [co-advisor with Zhiling Lan (UIC)]
- Matteo Del Grossi [M.S. 2023] (UIC) [co-advisor with Ian Kash (UIC)]
- Revathi Dhotre
- Boyang Li [Ph.D. 2023] (IIT) [co-advisor with Zhiling Lan (IIT)]
- Bharat Kale [M.S. 2018, Ph.D. 2023] (NIU) [co-advisor for Ph.D. with Maoyuan Sun (NIU)]
- Sreten Kljaic C² the P² Fellow
- Yuping Fan [Ph.D. 2021] (IIT) [co-advisor with Zhiling Lan (IIT)]
- Ryan Lewis [M.S. 2021] (NIU) [Advisor]
- Priyanjani Chandra [M.S. 2020] (NIU) [co-advisor with Pratoool Bharti (NIU)]
- Mrinal Roy [M.S. 2019] (NIU) [Advisor]
- Ankita Upadhyay [M.S. 2019] (NIU) [Advisor]
- Sean Wallace [Ph.D. 2017] (IIT) [co-advisor with Zhiling Lan (IIT)]
- Adam Young [M.S. 2015] (NIU) [Advisor]

- **Thesis and Dissertation Committees**

- Ahmedur Rahman Shovon [Ph.D. 2025] (UIC) [Dissertation]
- Noaman Ahmad [Ph.D. 2025] (UIC) [Dissertation]
- Giacomo Brunetta [M.S. 2025] (UIC) [Thesis]
- Niccolo Brembilla [M.S. 2025] (UIC) [Thesis]
- Valentino Guerrini [M.S. 2025] (UIC) [Thesis]
- Alessandro Febretti [Ph.D. 2017] (UIC) [Dissertation]
- Magda Baniukiewicz [M.S. 2017] (NIU) [Thesis]
- Joshua Boley [M.S. 2017] (NIU) [Thesis]
- Khairi Reda [Ph.D. 2014] (UIC) [Dissertation]

- **Past Posbacs, Predocs and Postdocs:**

- Aaron Knoll [Principal Research Scientist, AMD]
- Yash Kurkure (Predoc) [PhD student at University of Illinois Chicago]
- Wesley Kwiecinski (Postbac) [MS student at University of Illinois Chicago]
- Xinlian Liu [Deputy Scientific Information Officer, NICHD]
- Tommy Marrinan [Associate Professor, St. Thomas University]
- Eric Olson (Predoc) [Providence]
- Khairi Reda [Associate Professor, University of Illinois Chicago]
- Tushar Udeshi (Predoc) [Google, Inc.]
- Venkatram Vishwanath [Scientist, Argonne National Laboratory]
- Zhongkai Wen [co-supervised with Ian Kash (CS)]